Many Logics, One Methodology
— LogiKEy —

Christoph Benzmüller and Luca Pasetto

ESSLLI 2026

Slides are not final yet

Logical/Mathematical Imperialism

Choose some logical foundation and don't question them ever again.

For example, commit to a priori assumptions like:

$\exists x \,.\, x = x$	existence of objects, non-empty types
$\exists x \,.\, x = 1/0$	terms always denote, undefinedness exists
$\forall \varphi \,.\, \varphi \vee \neg \varphi$	law of excluded middle, classical logic
$\forall \varphi \psi \,.\, (\varphi \equiv \psi) \supset (\varphi = \psi)$	boolean extensionality
$\forall \varphi \psi \,.\, (\forall x \,.\, \varphi x = \psi x) \supset (\varphi = \psi)$	functional extensionality
If $H \vdash \varphi$ then $H, \psi \vdash \varphi$	monotonicity
$\ldots$	

Such choices are typically made in maths formalisation projects.

They might be completely unnaceptible for formalisations in philosophy, law, …

Logical/Mathematical Pluralism — LogiKEy

Make logical foundations flexible and negotiable.

Support reflection on them within proof assistants.

$\exists x \,.\, x = x$

$\exists x \,.\, x = 1/0$

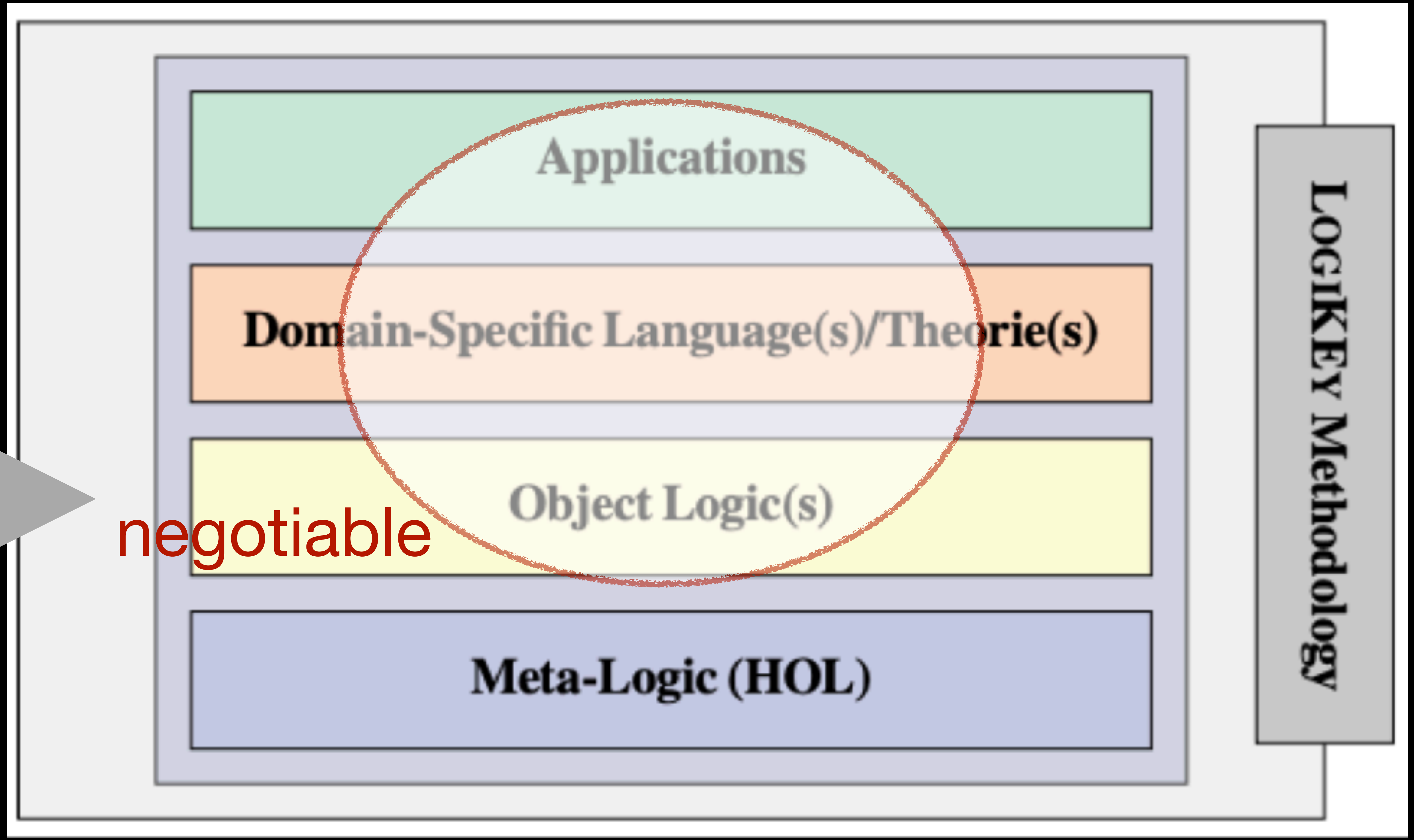
$\forall \varphi \,.\, \varphi \lor \neg \varphi$

$\forall \varphi \psi \,.\, (\varphi \equiv \psi) \supset (\varphi = \psi)$

$\forall \varphi \psi \,.\, (\forall x \,.\, \varphi x = \psi x) \supset (\varphi = \psi)$

If $H \vdash \varphi$ then $H, \psi \vdash \varphi$

$\ldots$



This way proof assistants may support formalisations also in philosophy, law, …

Logical/Mathematical Pluralism — LogiKEy

Make logical foundations flexible and negotiable.

Support reflection on them within proof assistants.

$\cancel{\exists x . x = x}$

$\cancel{\exists x . x = 1/0}$

Free Logic

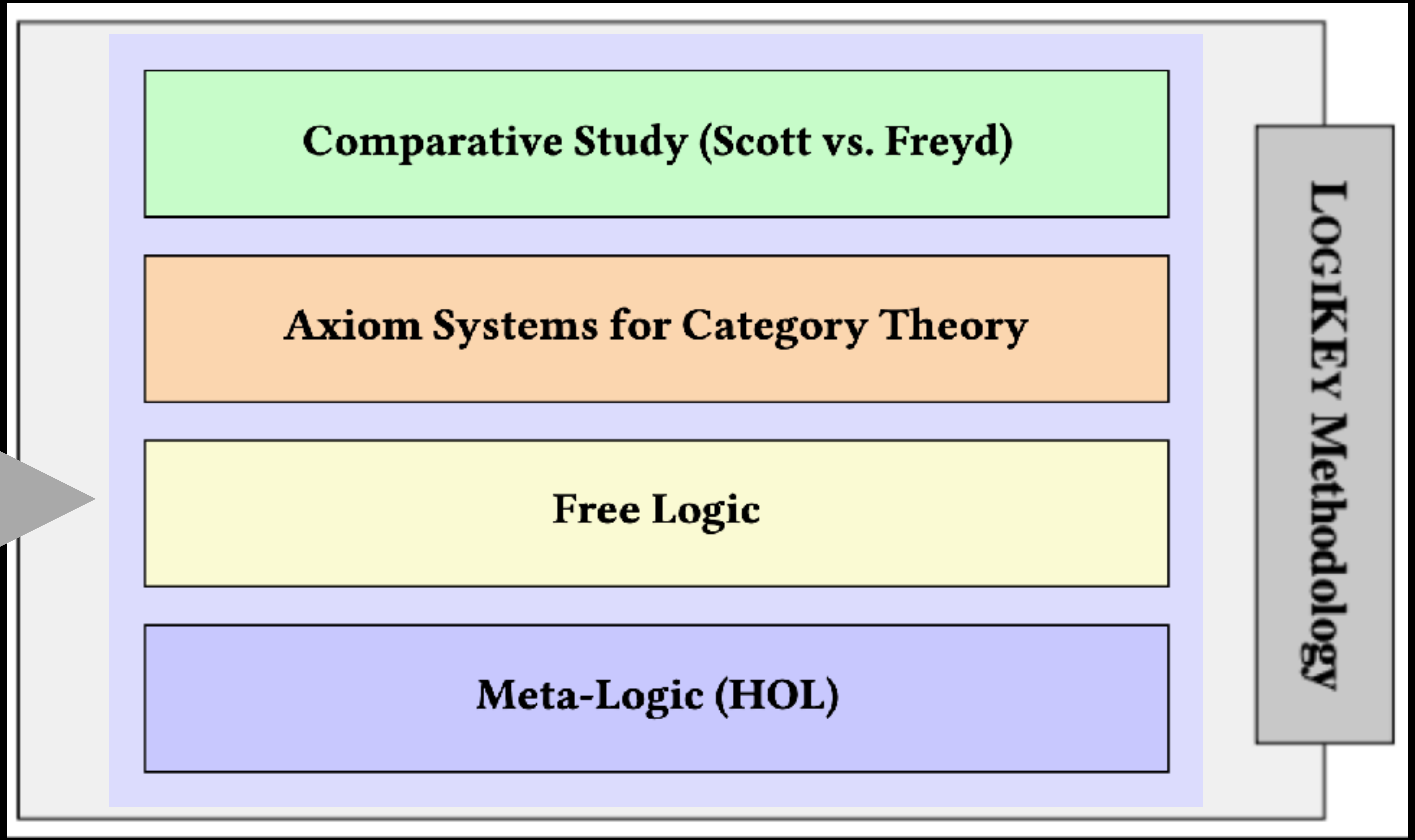
$\forall \varphi . \varphi \vee \neg \varphi$

$\forall \varphi \psi . (\varphi \equiv \psi) \supset (\varphi = \psi)$

$\forall \varphi \psi . (\forall x . \varphi x = \psi x) \supset (\varphi = \psi)$

If $H \vdash \varphi$ then $H, \psi \vdash \varphi$

$\ldots$



Logical/Mathematical Pluralism — LogiKEy

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Support reflection on them within proof assistants.

$\cancel{\exists x . x = x}$

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Free Logic

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If $H \vdash \varphi$ then $H, \psi \vdash \varphi$

$\dots$

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Automating Free Logic in HOL, with an Experimental Application in Category Theory

Published: 01 January 2019

Volume 64, pages 53–72, (2020) Cite this article

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Christoph Benzmüller & Dana S. Scott

Journal of Automated Reasoning

Journal of Automated Reasoning

Aims and scope →

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Category Theory in Isabelle/HOL as a Basis for Meta-logical Investigation

Conference paper | First Online: 28 August 2023

pp 69–83 | Cite this conference paper

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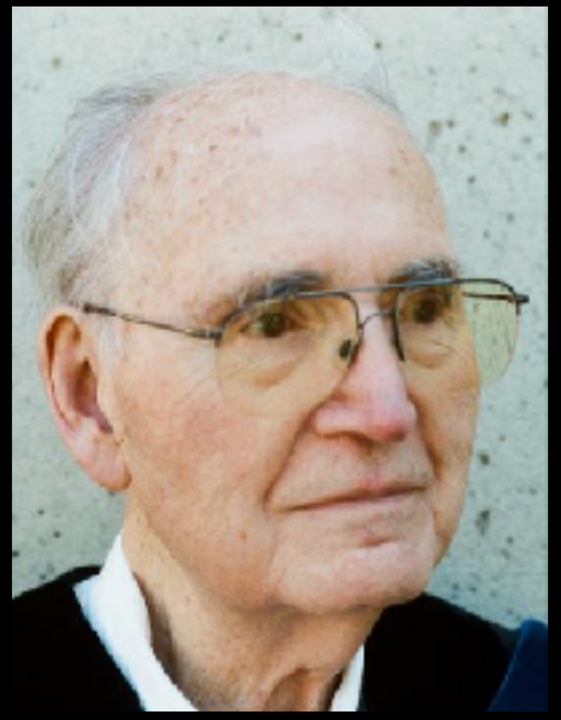
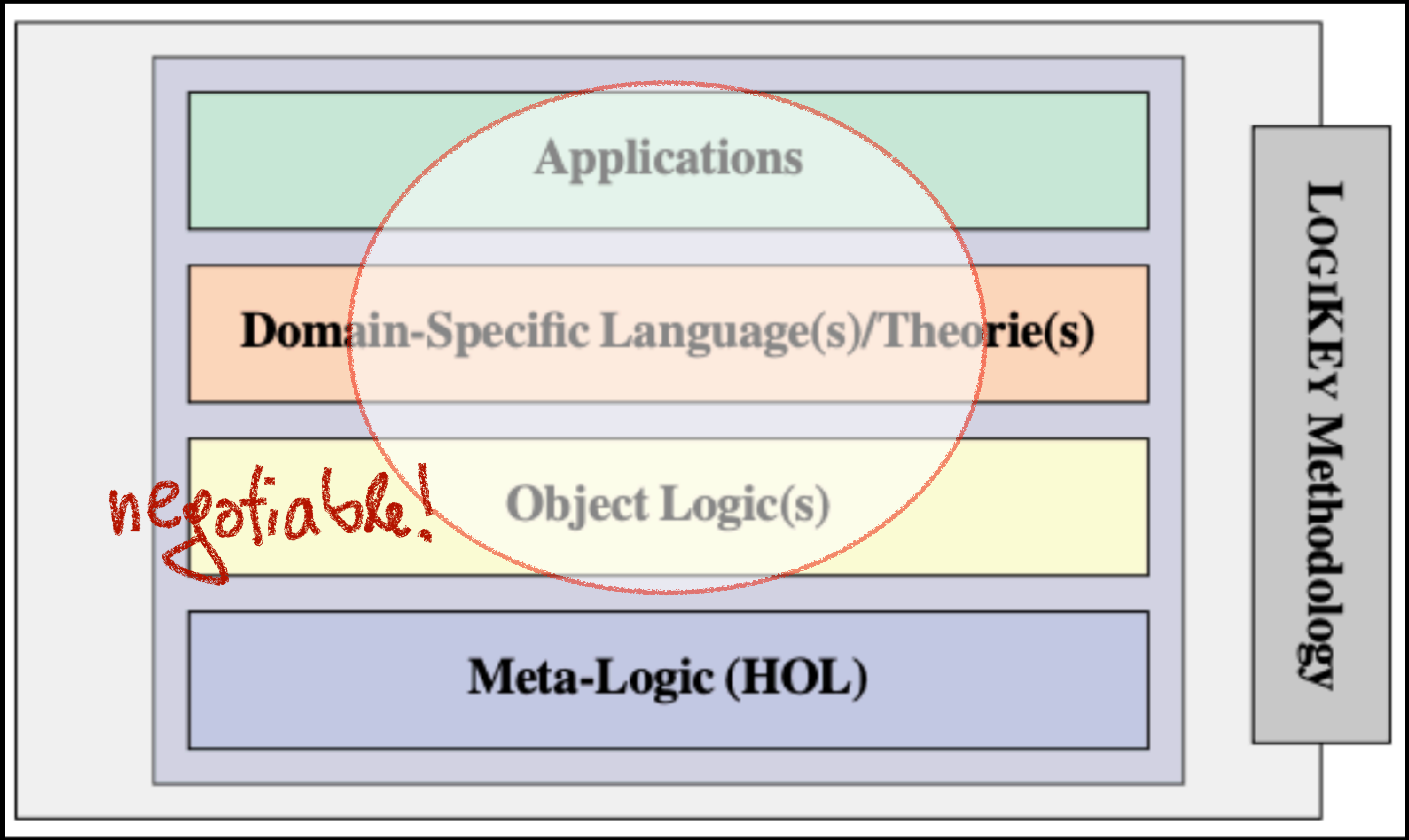
Intelligent Computer Mathematics (CICM 2023)

Jonas Bayer, Alexey Gonus, Christoph Benzmüller & Dana S. Scott



jww: **Dana Scott**

LogiKEy: Logic-pluralistic KR&R Methodology



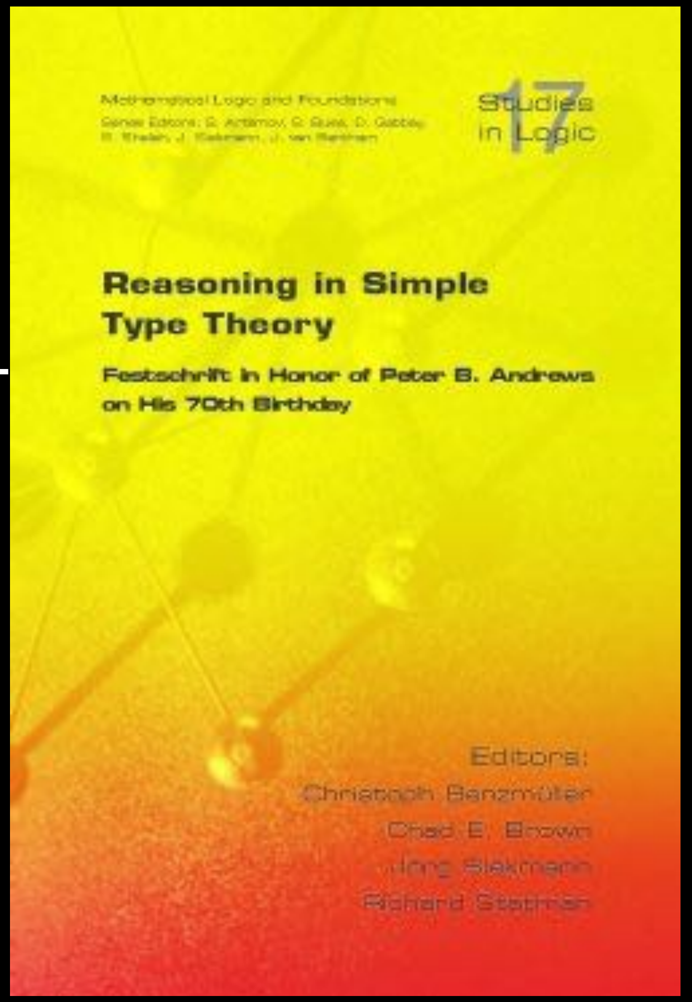
Origins: 2007-2008

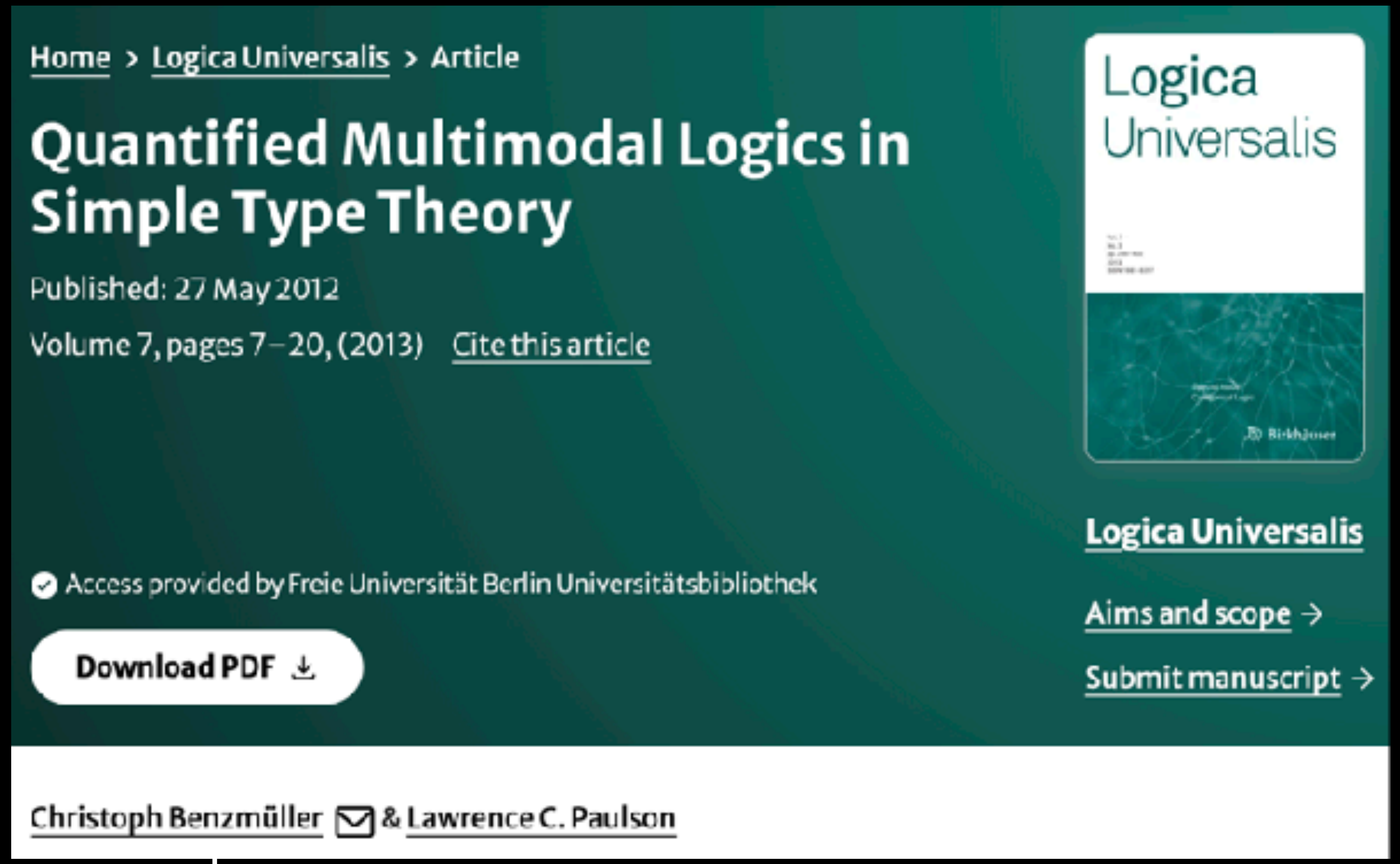
LogiKEy: Logic-pluralistic KR&R Methodology

Selection of logic embeddings:

- Multimodal Logics [Andrews Festschrift, 2008]
- Access Control Logic [SEP, 2009]
- Intuitionistic Logic [LJ IGPL, 2010]
- Quantified Multimodal Logic [Log Univers, 2012]
- Quantified Conditional Logic [J Phil Log, 2016]
- Multivalued Logic [LLP, 2016]
- Free Logic [JAR, 2020]
- Dyadic Deontic Logic [AIJ, 2020] [JANCL, 2024]
- (Hyper-)Intensional HO Modal Logic [RSL, 2020]
- Public Announcement Logic [JLC, 2022]
- Logic of Right to Know [ECAI, 2025]
- …

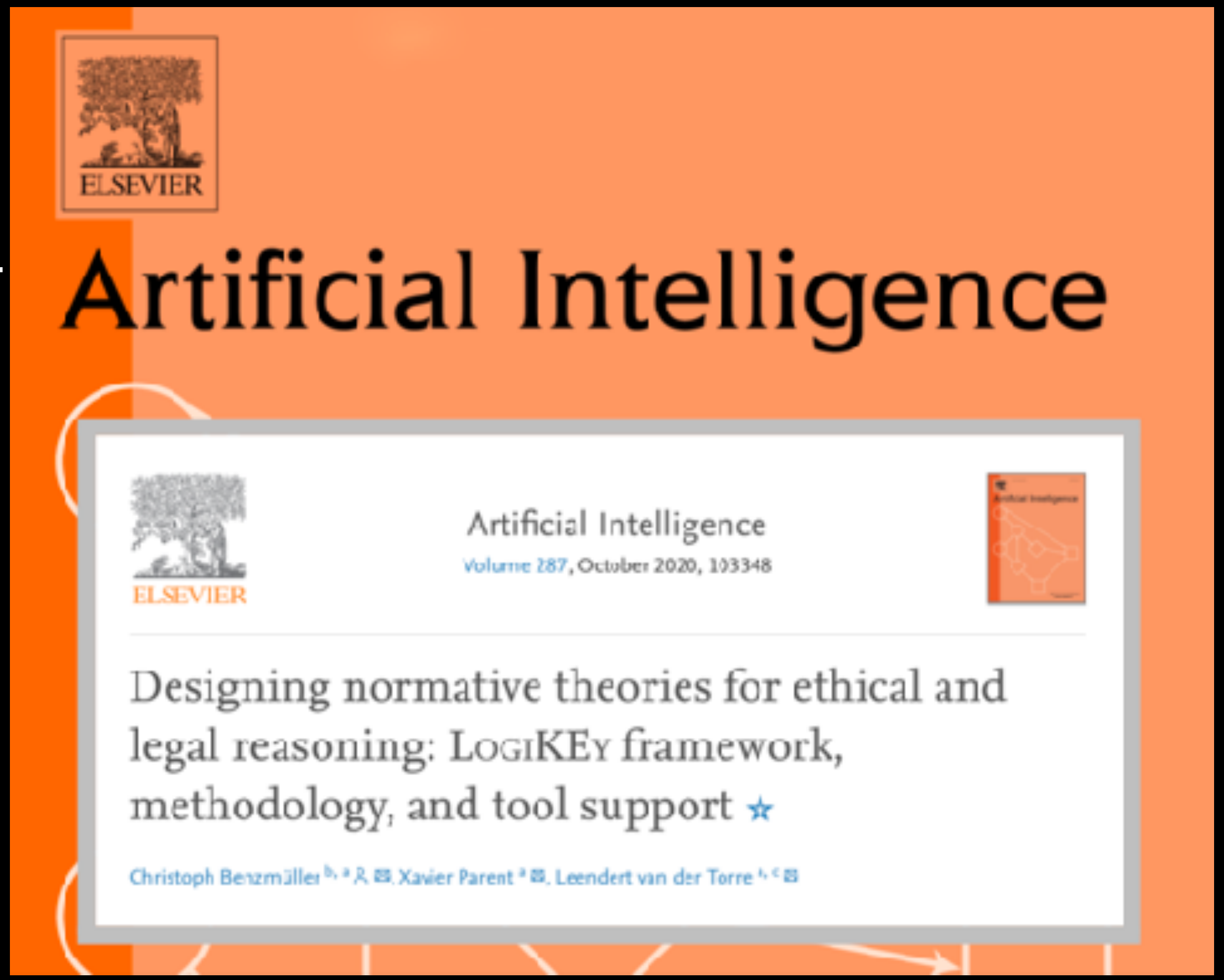
jww
Larry Paulson





jww: Larry Paulson

L. van der Torre & X. Parent
jww



Term
LogiKEy
introduced
in 2020

LogiKEy: Logic-pluralistic KR&R Methodology

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Notes on Gödel's and Scott's variants of the ontological argument

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Christoph Benzmüller & Dana Scott

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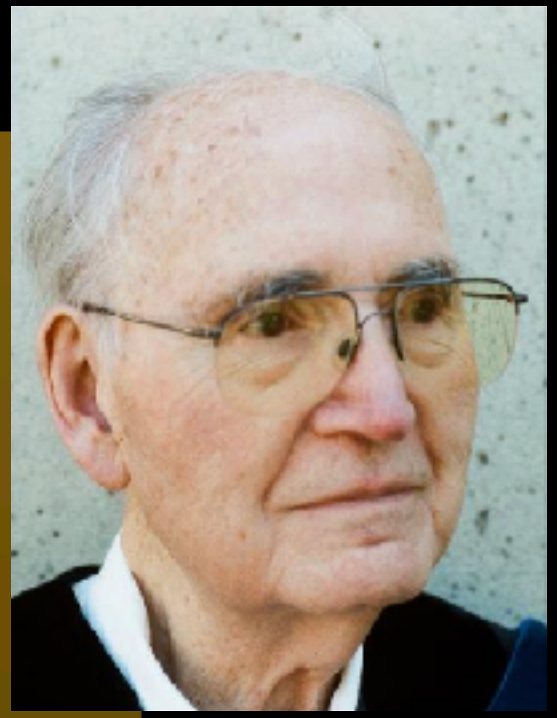
Abstract

Notes on Kurt Gödel's modal ontological argument and Dana Scott's variant of it are presented. These remarks, supported by experimental studies with a proof assistant system for classical higher-order logic, implicitly answer some questions the authors have received over the last decade(s). In addition, some new insights resulting from the conducted experiments are reported.

2025



jww: Dana Scott

LogiKEy@work in Computational Metaphysics

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Formel von Kurt Gödel: Mathematiker bestätigen Gottesbeweis

Von Tobias Hürter

picture-alliance/ Imagno/ Wiener Stadt- und Landesbibliothek

Kurt Gödel (um das Jahr 1935): Der Mathematiker hielt seinen Gottesbeweis jahrzehntelang geheim

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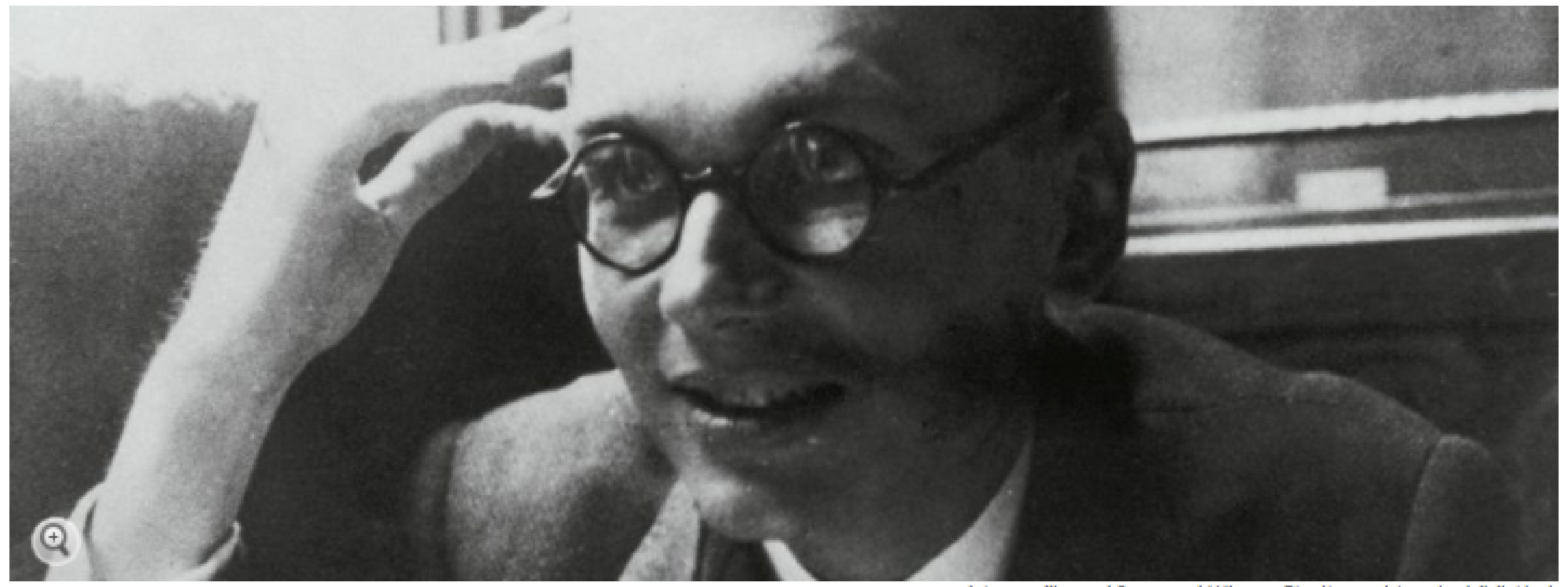
English Site > Germany > Science > Scientists Use Computer to Mathematically Prove Gödel God Theorem

Holy Logic: Computer Scientists 'Prove' God Exists

By David Knight



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Austrian mathematician Kurt Gödel kept his proof of God's existence a secret for decades. Now two scientists say they have proven it mathematically using a computer.

Two scientists have formalized a theorem regarding the existence of God penned by mathematician Kurt Gödel. But the God angle is somewhat of a red herring -- the real step forward is the example it sets of how computers can make scientific progress simpler.

2013

Bruno Woltzenlogel-Paleo

Formel von Kurt Gödel: Mathematiker bestätigen Gottesbeweis

Von Tobias Hürter

Holy Logic: Computer Scientists 'Prove' God Exists

By David Knight

MEDIA & CULTURE

Is God Real? Scientists 'Prove' His Existence With Godel's Theory And MacBooks

HOME / SCIENCE NEWS

Researchers say they used MacBook to prove Goedel's God theorem

God exists, say Apple fanboy scientists

With the help of just one MacBook, two Germans formalize a theorem that confirms the existence of God.

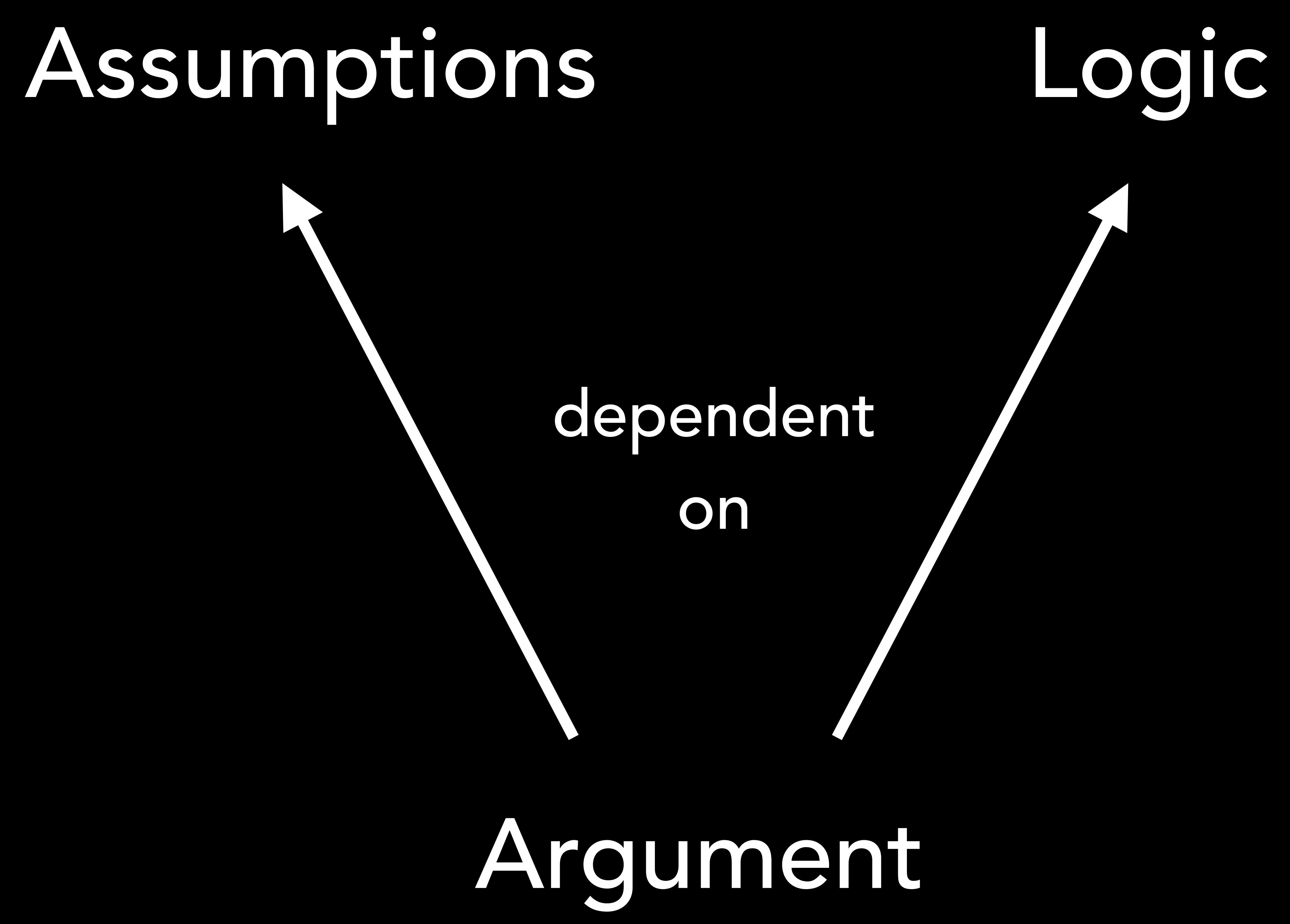
Chris Matyszek, c|net

Important Clarification

No!

We haven't proven the existence of God using a computer!

We have formally analysed a prominent argument for the existence of God using a computer.



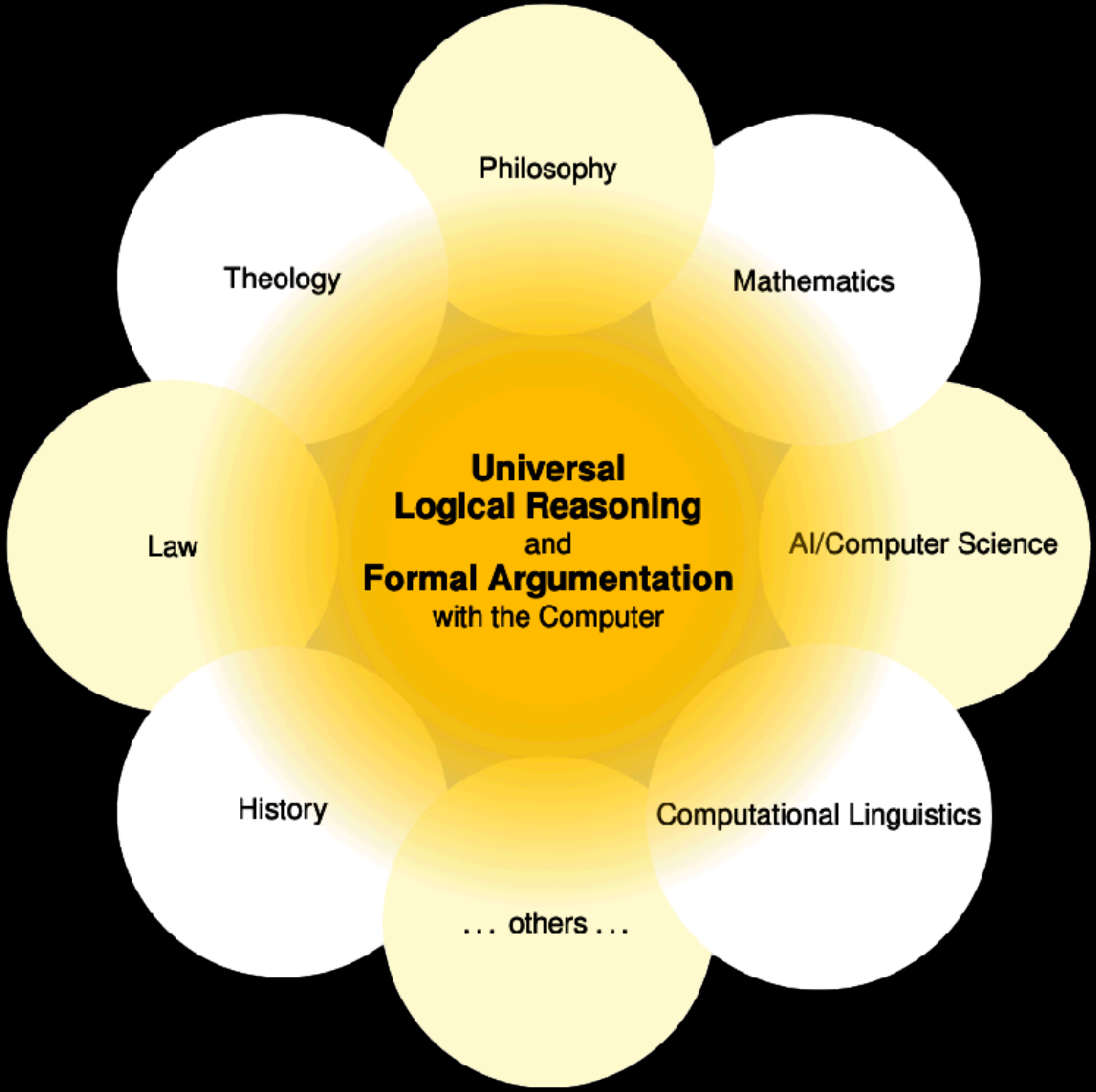
‘Exact’ science is possible …

Es gibt syst<ematische> Methoden zur Lösung aller Probl<eme> (auch Kunst etc.).

Es gibt eine wissenschaftliche (exakte) Ph<ilosophie> {und Th<eologie>}, welche die Begriffe der höchsten Abstraktheit behandeln.

Kurt Gödel: Maximen V

Kurt Gödel (1906-78)



… far beyond the realm of mathematics

LogiKEy can be used with many different tools

Isabelle/HOL: Still my favorite tool for logic education and research

- Proof automation with **Sledgehammer** (FO & HO ATPs, SMT solvers)
- Model- and countermodel-finding with **Nitpick**
- Powerful **User Interface** (but this is a bottleneck at the same time)

Many other provers can be used instead (and we have demonstrated this):

- Other ITPs: **Lean, HOL4, Coq, Agda, PVS, …**
- Own HO ATPs: **LEO-II, LEO-III**
- Other HO-ATPs: **Vampire, E, Zipperpin, CVC5, …**